

Trong Nhan Nguyen

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EDUCATION

Pukyong National University, Busan 48513, KOREA

Master student, Advisor: (Prof.) [Junghwan Oh](#)

Oct. 2023 – Oct. 2025

Topic 1: Non-Destructive Testing (Scanning Acoustic Microscopy system)

Topic 2: Research and apply methodologies to address challenges in low-level vision.

Topic 3: Inspection system

Artificial Intelligence Center (AIC), FPT Software, Ho Chi Minh City, VIETNAM

Solution Department, Director: (Dr.) [Phong X. Nguyen](#)

Oct. 2022 - Dec. 2023

AI Camp Central Talent Academy - FPT Software, Quy Nhon City, VIETNAM

Memtee Batch 7

Mar.2022 - Oct. 2022

Ho Chi Minh of Technology University, Thu Duc City, VIETNAM

B.S. in Information Technology Engineering, Advisor: (Prof.) [Hung Nguyen](#)

Aug.2018 - Dec. 2022

PROFESSIONAL SUMMARY

- Experience in Data processing, Data Analysis using Python.
- Experience with NDT (Non Destructive Testing) inspection system ([Scanning Acoustic Microscopy](#)).
- Experience with Laser (Micro Epsilon), Camera (Basler) inspection.
- Knowledge of Machine Learning, Deep Learning, specialized in Computer Vision.
- Experience in deploying AI model to Robotic system (Yaskawa).
- Experience in implementing and developing web frameworks using Python.
- Experience in various industrial domains, including Air Handling Unit systems, steel manufacturing processes, medicine (search engine optimization), and industrial inspection (camera inspection, laser inspection, and NDT inspection).

RESEARCH INTERESTS

- Low-level vision: super resolution, image denoising, image inpainting.
- Developing segmentation models to detect defects.
- Exploratory Data Analysis with multi domain (*i.e.*, manufacturing process, industrial inspection, material).
- Applied AI models to automate inspection systems in industrial settings.
- Integrated and interacted with robotic arms, cameras, lasers, and 3D scanners.
- Technique: Generative Adversarial Network, Transformer, Diffusion model, Mamba.

RESEARCH AND ACADEMIC EXPERIENCE

Graduate Research Assistant

Pukyong National University, Busan, Korea

Oct. 2023 - present

1. *Development of the robotic arm system to automatic identify, scan and inspect the surface of Weld Steam*

Objective: The Welding Auto Inspection project aims to inspect the surface of welds using a Micro Epsilon 30x2-50 laser system combined with a Basler camera for automated weld identification. The project has successfully integrated multiple modules, resulting in a fully automated inspection process.

- Laser Scanning Performance: Scanning speed of 65 mm/s with a detailed resolution of 100 μm x 49 μm .
- High Accuracy: The Vision module attained 95.36% accuracy in weld identification
- Software Enhancement: The new control system control robot individually, enabling real-time control and more accurate robot behavior tracking.
- Rapid 3D Visualization: Processing time under 1 second (approximately 0.5 seconds).

- Comprehensive Automation: With Flat Object, all modules from Vision, Laser, Software to Robot and Inspection have been completed, creating a fully automated inspection process.
2. *Development of AI Software to inspect defect for Scanning Acoustic Microscopy software*
 - Build Deep Learning model to detect some common defect on power semiconductor
 - Develop the principle security for model
 - The output is the name of defect and to calculate the percentage defect of sample.
 3. *Develop MLCC Inspection system, OHLABS company*
 - Build AI Flow to define the abnormal MLCC.
 - Laser Marking stage to mark the abnormal MLCC.

Undergraduate Research Assistant Ho Chi Minh University of Technology, Vietnam Feb. 2021- Mar. 2022

1. *Vietnamese Chatbot And OCR For Paperwork*

- The application has two main features, including OCR to extract personal information and then automatically fill it into paper and a chatbot to answer questions.

INDUSTRIAL EXPERIENCE

AI Engineer Artificial Intelligence Center, FPT Software, Vietnam Oct. 2022 - Dec. 2023

1. *Project : Use machine learning to optimize the dehumidification process by supplying optimal air temperature to gallery air handlers of Chicago Art Museum ([Ameresco](#) company).*
 - Data Analysis for Museum 1, Museum 2, Museum 3, Museum 4.
 - Define and implement algorithms to calculate heat gain for the space temperature when have change CO2 level.
 - Model forecasting CO2 level and clustering.
 - Optimize algorithm.
2. *Project : Use machine learning to optimize defect detection in the steel production process ([Yamato](#) company).*
3. *Project : Development Deep Learning Process to enhance search engine ([Long Chau Retail](#) company).*
 - Data Analysis for Medicine
 - Combine between Full-text search and Semantic search.
 - Deploy DL model to Long Chau Retail's website.

AI Engineer Smart City department, FPT Information System, Vietnam Jan. 2022- Mar. 2022

- Detect and warn the center management system about encroachment on the sidewalk.
- Model will be deployed into camera.
- Use camera system of Traffic Management at Ho Chi Minh City.

RESEARCH PUBLICATIONS [Google scholar: Trong Nhan Nguyen](#)

(*First Author, **Co-First Author and ***Co-author)

Journal Publications

1. **Trong Nhan Nguyen***, Vu Hoang Minh Doan, Tan Hung Vo, Dang Khoa Pham, Quoc Dung Nguyen, Truong Tien Vo, Jaeyeop Choi, Junghwan Oh. "Unsupervised Hybrid Transformer-Mamba Diffusion for Depth-Dependent Image Denoising in Scanning Acoustic Microscopy", *IEEE Transactions on Industrial Informatics*, IF = 9.9, 2026. [10.1109/TII.2025.3648838](#)
2. **Trong Nhan Nguyen***, Vu Hoang Minh Doan, Tan Hung Vo, Jaeyeop Choi, Junghwan Oh. "Transformer-Based Super-Resolution Scanning Acoustic Imaging for Industrial Inspection", *IEEE International Conference on Industrial Technology (ICIT 2025)*. 2025. <https://ieeexplore.ieee.org/document/10965207>
3. **Trong Nhan Nguyen***, Vu Hoang Minh Doan, Tan Hung Vo, Dinh Dat Vu, Le Hai Tran, Thi Thu Ha Vu, Truong Tien Vo, Quoc Dung Nguyen, Sudip Mondal, Jaeyeop Choi, Byeongil Lee, Junghwan Oh. "Image-

enhanced Generative Models for Industrial Scanning Acoustic Super-resolution”, *Engineering Applications of Artificial Intelligence*, *IF* = 8.0, 2025. <https://doi.org/10.1016/j.engappai.2025.112500>

4. Thi Thu Ha Vu, Tan Hung Vo, **Trong Nhan Nguyen****, Jaeyeop Choi, Le Hai Tran, Vu Hoang Minh Doan, Van Bang Nguyen, Wonjo Lee, Sudip Mondal and Junghwan Oh. “GAN-Based Super-Resolution in Linear R-SAM Imaging for Enhanced Non-Destructive Semiconductor Measurement”. *Applied Sciences*, *IF* = 2.5, 2025. <https://www.mdpi.com/2076-3417/15/12/6780>
5. Truong Tien Vo, Quy Phuong Le, **Trong Nhan Nguyen*****, Sudip Mondal, Jaeyeop Choi, Byeongil Lee, Junghwan Oh. “A Non-Contact Ballistocardiogram-Based System with Multi-Task Deep Learning for Real-Time Vital Signs Monitoring in Acupuncture”, *Computers in Biology and Medicine*, *IF* = 6.3, 2026. <https://doi.org/10.1016/j.combiomed.2026.111461>
6. Thi Thu Ha Vu, Tan Hung Vo, **Trong Nhan Nguyen*****, Jaeyeop Choi, Sudip Mondal, Junghwan Oh. “Optimizing Scanning Acoustic Tomography Image Segmentation With Segment Anything Model for Semiconductor Devices”. *IEEE Transactions on Semiconductor Manufacturing*, *IF* = 2.3, 2024. <https://ieeexplore.ieee.org/document/10645767>

Manuscripts under review

1. **Trong Nhan Nguyen***, Vu Hoang Minh Doan, Tan Hung Vo, Jaeyeop Choi, Junghwan Oh. “Scanning Acoustic Image Restoration Using Diffusion for Automatic Reliable Inspection”, *Computers and Electrical Engineering*, *IF* = 4.9, Under Review.
2. Truong Tien Vo, Huu Sang Nguyen, **Trong Nhan Nguyen*****, Le Hai Tran, Quy Phuong Le, Tan Tien Nguyen, Jaeyeop Choi and Junghwan Oh. “Multimodal Smart Clothing with Haptic Feedback for Real-Time Muscle Activation Assessment in Self-Coaching Fitness”, *IEEE Internet of Things*, *IF* = 8.9, Revision.
3. Minh Ngoc Nguyen, Khai Le-Duc, Tan-Hanh Pham, **Trong Nhan Nguyen*****, Trang Nguyen, Quang Minh Luu, Ba Kien Tran, Truong-Son Hy, Viktor Dremine, Sergei Sokolovsky, Edik Rafailov. “A Wearable Device Dataset for Mental Health Assessment Using Laser Doppler Flowmetry and Fluorescence Spectroscopy Sensors”, *Communications Medicine - Nature*, *IF* = 6.3, Minor Revision, <https://arxiv.org/abs/2502.00973>.

Manuscripts submitted

1. Quoc Dung Nguyen, Le Hai Tran, Tan Hung Vo, Dinh Dat Vu, Thi Thu Ha Vu, **Trong Nhan Nguyen*****, Truong Tien Vo, Vu Hoang Minh Doan, Van Hiep Pham, Jaeyeop Choi, Sudip Mondal and Junghwan Oh. “Dual-Fast Scanning Acoustic Microscopy (DFSAM): A High-Speed Dual-Side Ultrasound Imaging System for Thick Sample Inspection”, *Applied Acoustics*, *IF* = 3.4.

Manuscript preparation

1. **Trong Nhan Nguyen***, Truong Son Hy, Junghwan Oh, “WaFIR: Wavelet-Fourier Image Restoration via Prompt-Guided Spectral Mixture-of-Experts”, *AAAI Conference* 2027

HONORS, AWARDS, AND ACHIEVEMENTS

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- Blue Scholarship, Brain Korea 21(BK21), Korean Government Scholarship Program, 2025
 - Fire Grant Scholarship, 2025
 - Scholarship, Pukyong National University, 2023
 - Impressive Award in the HUTECH Start-up Wing, 2021
 - Consolation Prize in the AI Hackerthon Contest, Ho Chi Minh University of Technology, 2020
 - Rewards of Undergraduate Research, Ho Chi Minh University of Technology, 2020

- Outstanding Undergraduate Student Scholarships, Ho Chi Minh University of Technology, 2018

SERVICES

Peer Reviewer: *IEEE Sensors; Journal of Infrastructure, Policy and Development; IEEE International Conference on Industrial Technology, Machine Vision and Applications; Annals of Mathematics and Physics, Discover Artificial Intelligence, Discover Sensors*